Appendix: Key Formulas

CMOS Inverter

Triode region: $V_{GS} > V_T$ and $V_{DS} < V_{GS} - V_T$; device behaves as a voltage-controlled resistor with a continuous channel.

Saturation region: $V_{GS} > V_T$ and $V_{DS} \geq V_{GS} - V_T$; channel is pinched off at the drain and I_D is primarily controlled by V_{GS}.

Logic HIGH input ($V_{in} = V_{DD}$):	nMOS ON (pull-down), pMOS OFF; $V_{out} \approx 0$
Logic LOW input ($V_{in} = 0$):	nMOS OFF, pMOS ON (pull-up); $V_{out} \approx V_{DD}$
Static power dissipation:	$P_{\text{static}} = I_{\text{static}} \times V_{DD} \approx 0$ (ideal, one transistor always OFF)

Photonics and Laser Principles

Photon energy:	$E = h\nu = \dfrac{hc}{\lambda}$
Wavelength from bandgap:	$\lambda_{\text{peak}} \approx \dfrac{hc}{E_g} = \dfrac{1.240\ \text{eV} \cdot \mu\text{m}}{E_g\ (\text{eV})}$
Absorption:	Photon absorbed $\rightarrow$ electron promoted from valence to conduction band
Spontaneous emission:	Electron falls from conduction to valence band $\rightarrow$ random photon emitted
Stimulated emission:	Incoming photon triggers electron transition $\rightarrow$ identical photon emitted (same phase, frequency, direction, and polarization as incoming photon)

Laser Oscillation Condition (Feedback Loop Analogy)

Closed-loop transfer function:	$\dfrac{V_{\text{out}}}{V_{\text{in}}} = \dfrac{A}{1 - A\beta}$
Barkhausen criterion for oscillation:	$\lvert A\beta \rvert = 1$ and $\angle A\beta = 0^\circ$ (or $360^\circ \times n$)
Physical meaning:	Round-trip gain = 1, total round-trip phase shift = 0 (or multiples of 2π)
Laser threshold condition:	Gain medium amplification = Mirror losses + Material losses (Equivalent to $\lvert A\beta \rvert = 1$ in the optical domain)

Population Inversion

For lasing:	Population inversion required: more electrons in excited state than ground state
Achieved by:	Optical or electrical pumping to create nonequilibrium carrier distribution